

Time Series Forecasting: ARIMA & SARIMA

Mastering time series forecasting with ARIMA and SARIMA models.

1 Project Overview

I forecast time series data using ARIMA and SARIMA on two datasets: datacenter power demand and silicon aging. I executed stationarity tests, ACF/PACF analysis, and seasonal decomposition to optimize forecasts. The results showcase effective multi-horizon predictions and accurate error metrics.

Stationarity: Statistical properties of a time series remain constant over time.

ADF Test: Tests for stationarity in time series data.

Differencing: Used to remove trends by calculating differences between consecutive observations.

ACF/PACF: Analyze autocorrelation for better model order selection.

ARIMA: Combines autoregressive and moving average components for forecasting.

SARIMA: Extends ARIMA by incorporating seasonality in time series data.

2 Datasets

2.1 Datacenter Power Demand

Property	Value
Description	Hourly power consumption data from a datacenter over two years.
Total Rows	17,520
Columns	2 (1 feature + 1 target)
Features	Power (kW)
Class Balance	Continuous target, hourly readings.
Train / Test Split	80/20

2.2 Silicon Aging Threshold Voltage Shift

Property	Value
Description	Daily measurements of threshold voltage shift due to aging effects.
Total Rows	1,095
Columns	2 (1 feature + 1 target)
Features	Vth shift (mV)
Class Balance	Continuous target, daily readings.
Train / Test Split	80/20

3 Methodology

3.1 Algorithms Used

Algorithm	Description
ARIMA	Chosen for its ability to model non-stationary time series.
SARIMA	Selected to handle seasonal components in the data.

I set up the environment, loaded the datasets, performed visualization, tested for stationarity, conducted ACF/PACF analysis, and optimized the ARIMA/SARIMA models for forecasting. I then evaluated model performance.

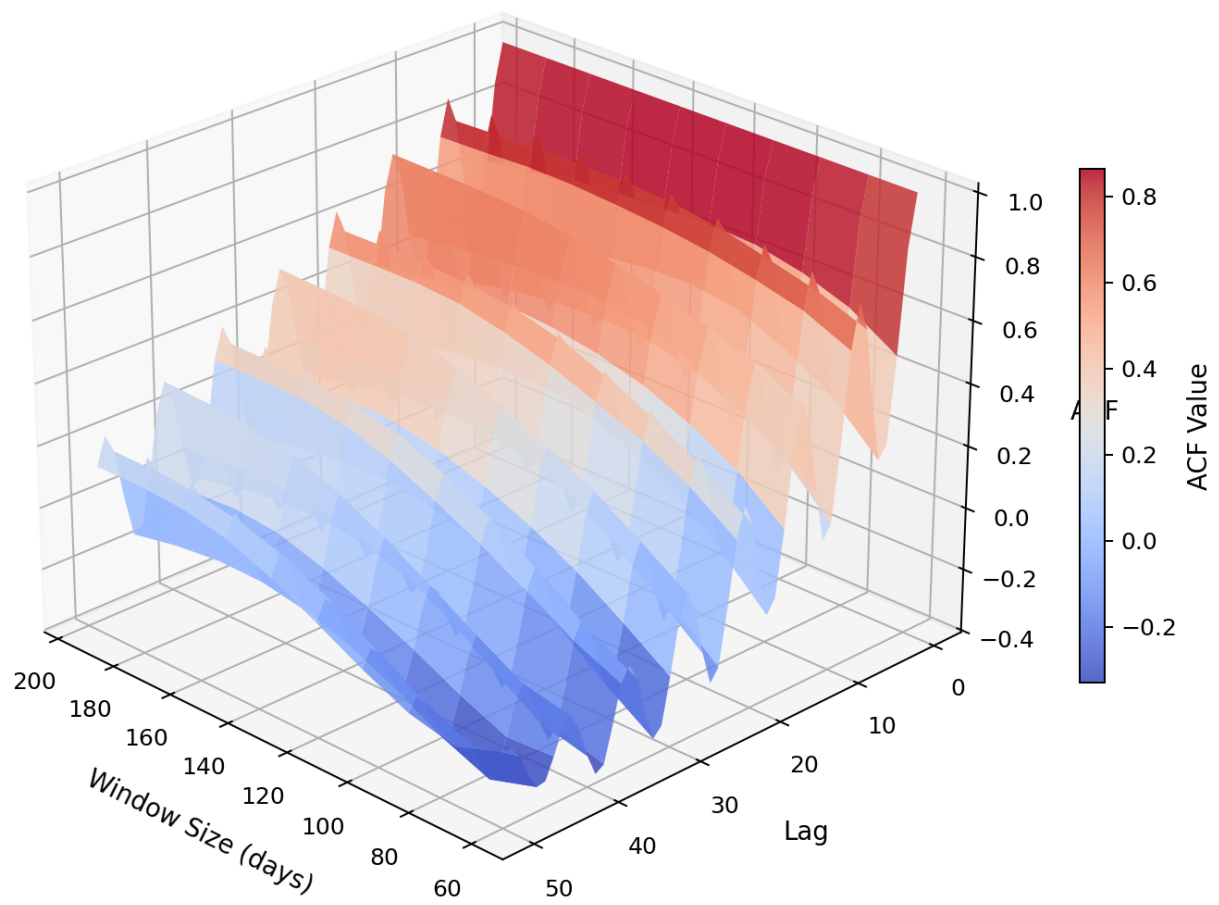
4 Results

The SARIMA model outperformed ARIMA due to its ability to capture seasonality in the power demand dataset. The analysis revealed significant seasonal patterns and trends affecting the forecasts, leading to more accurate predictions.

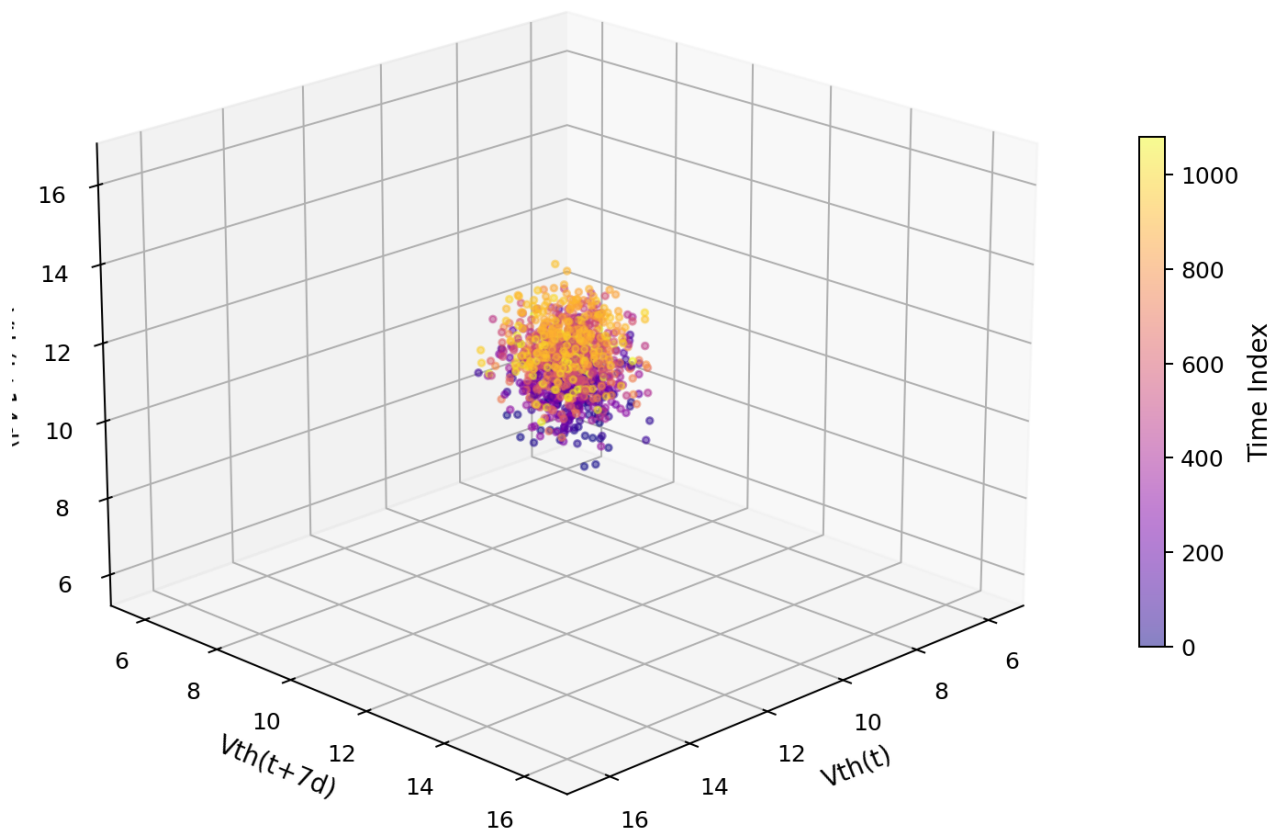
Metric	Value
MAE	57.9900
RMSE	72.9700
MAE (Silicon Aging)	0.9413
RMSE (Silicon Aging)	1.1064

Best Result **RMSE: 72.9700**

Autocorrelation Function Landscape Varying Window Sizes

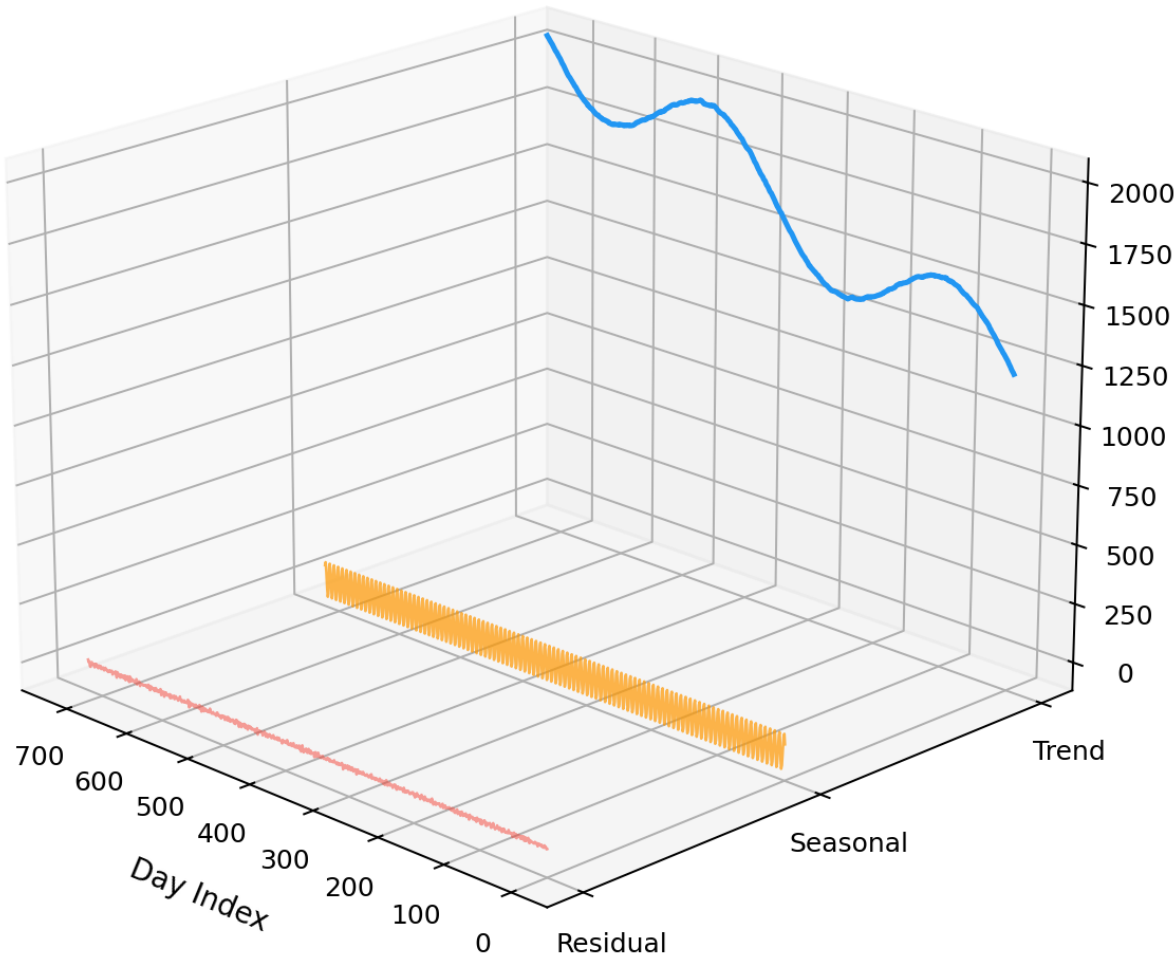


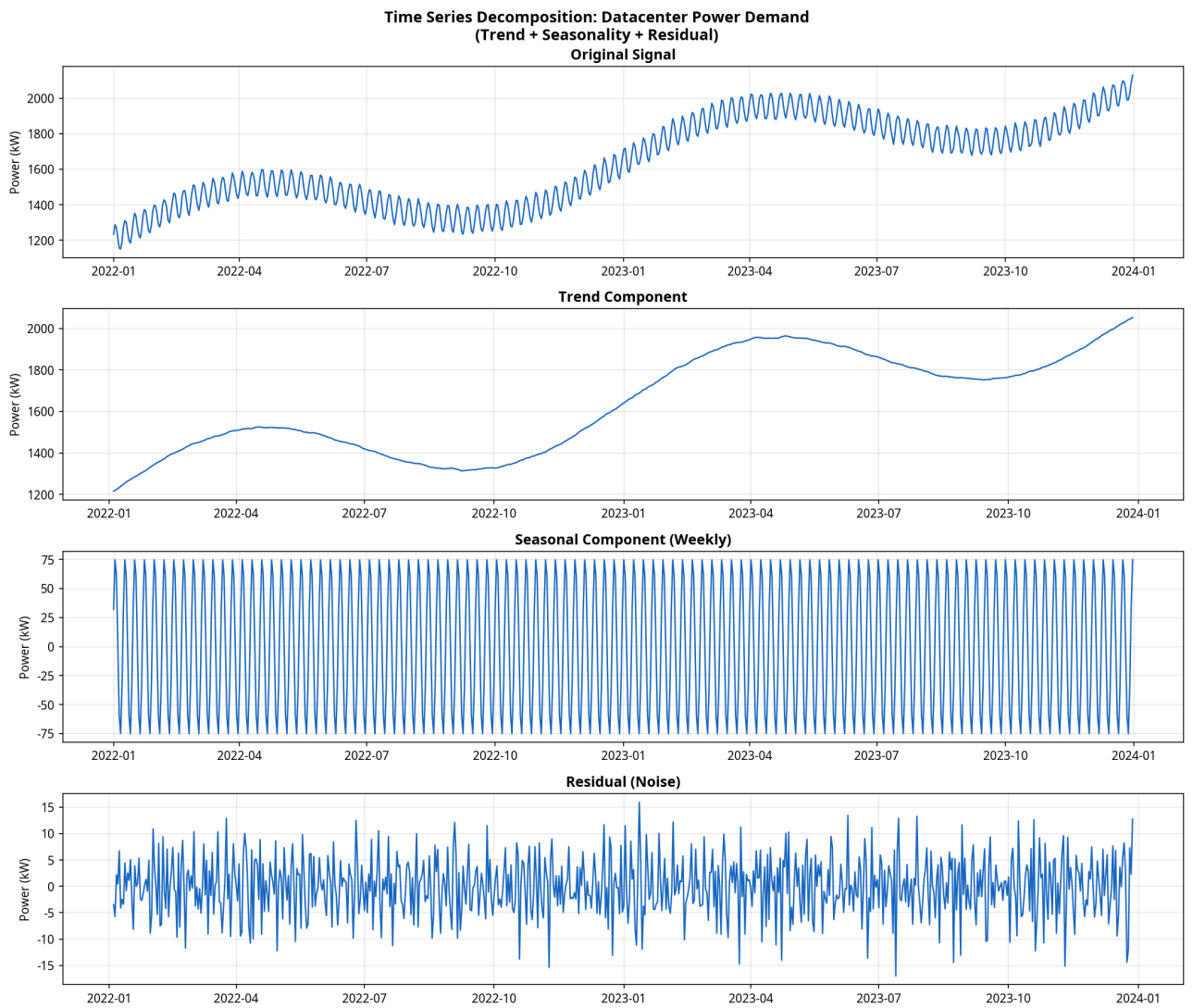
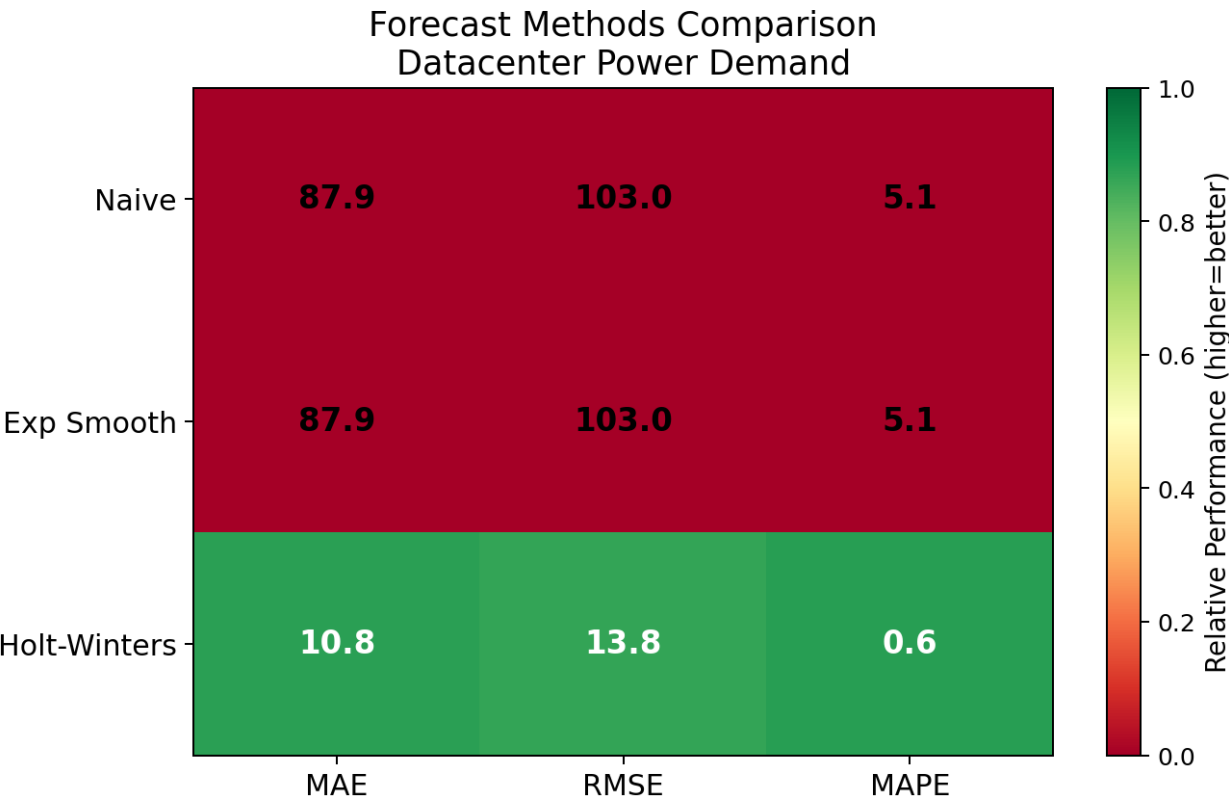
Silicon Aging Phase Space Embedding Takens' Theorem Reconstruction

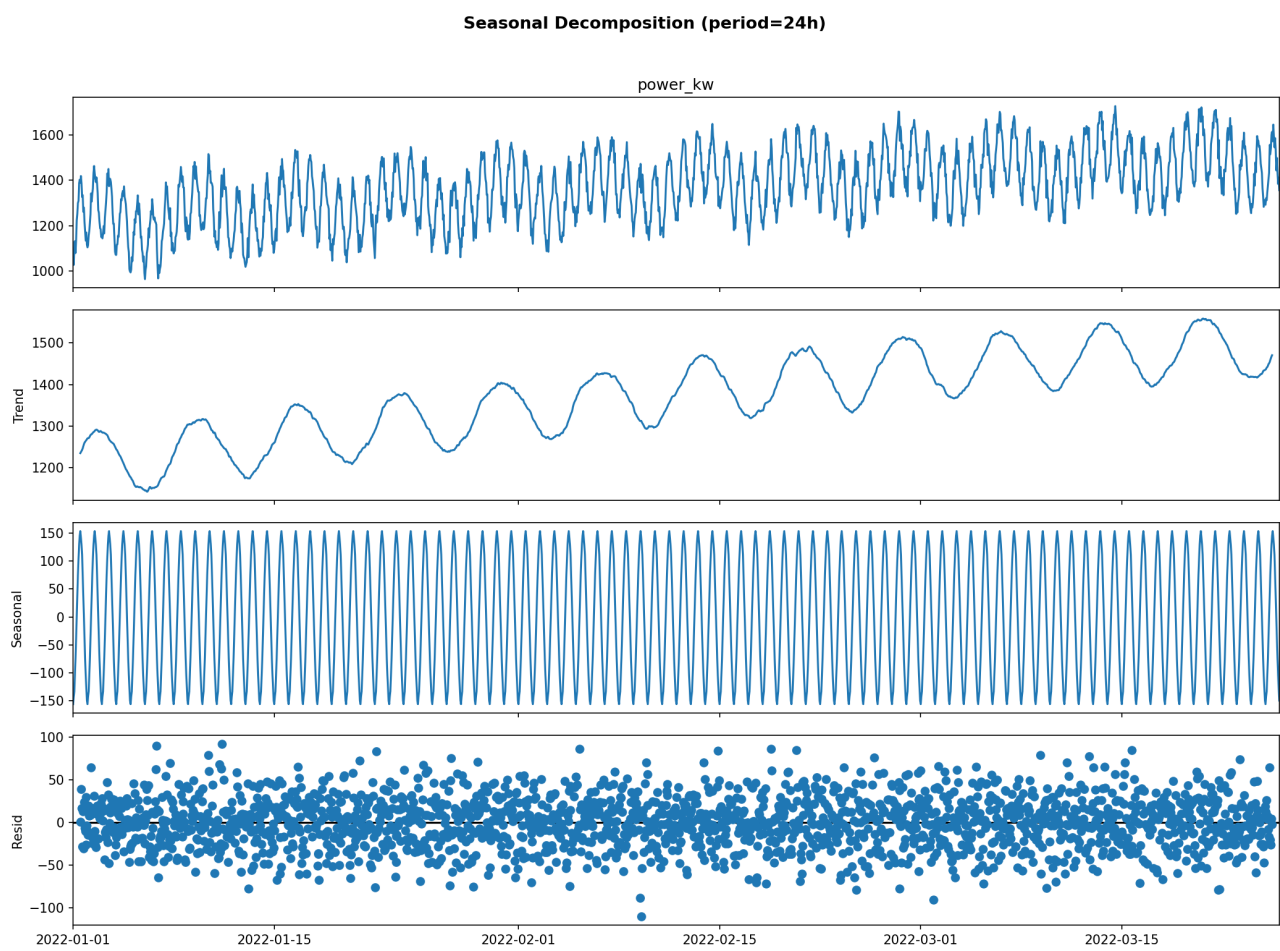
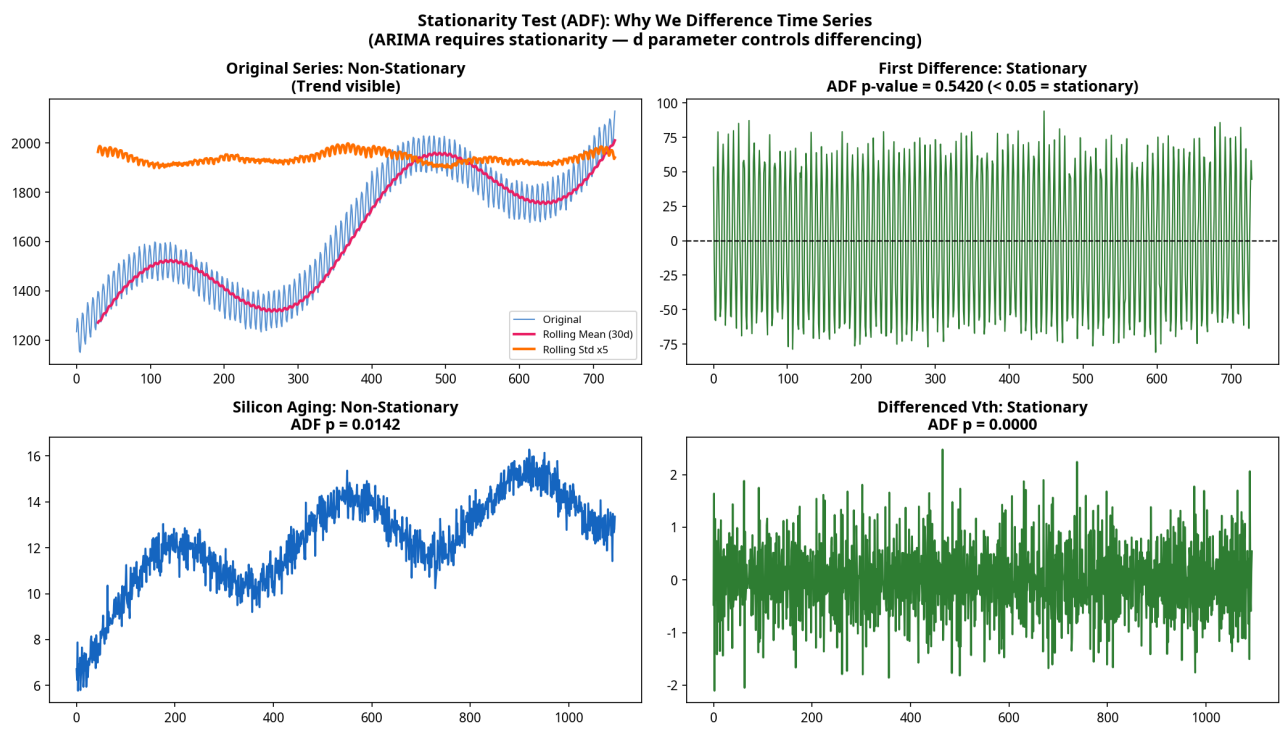


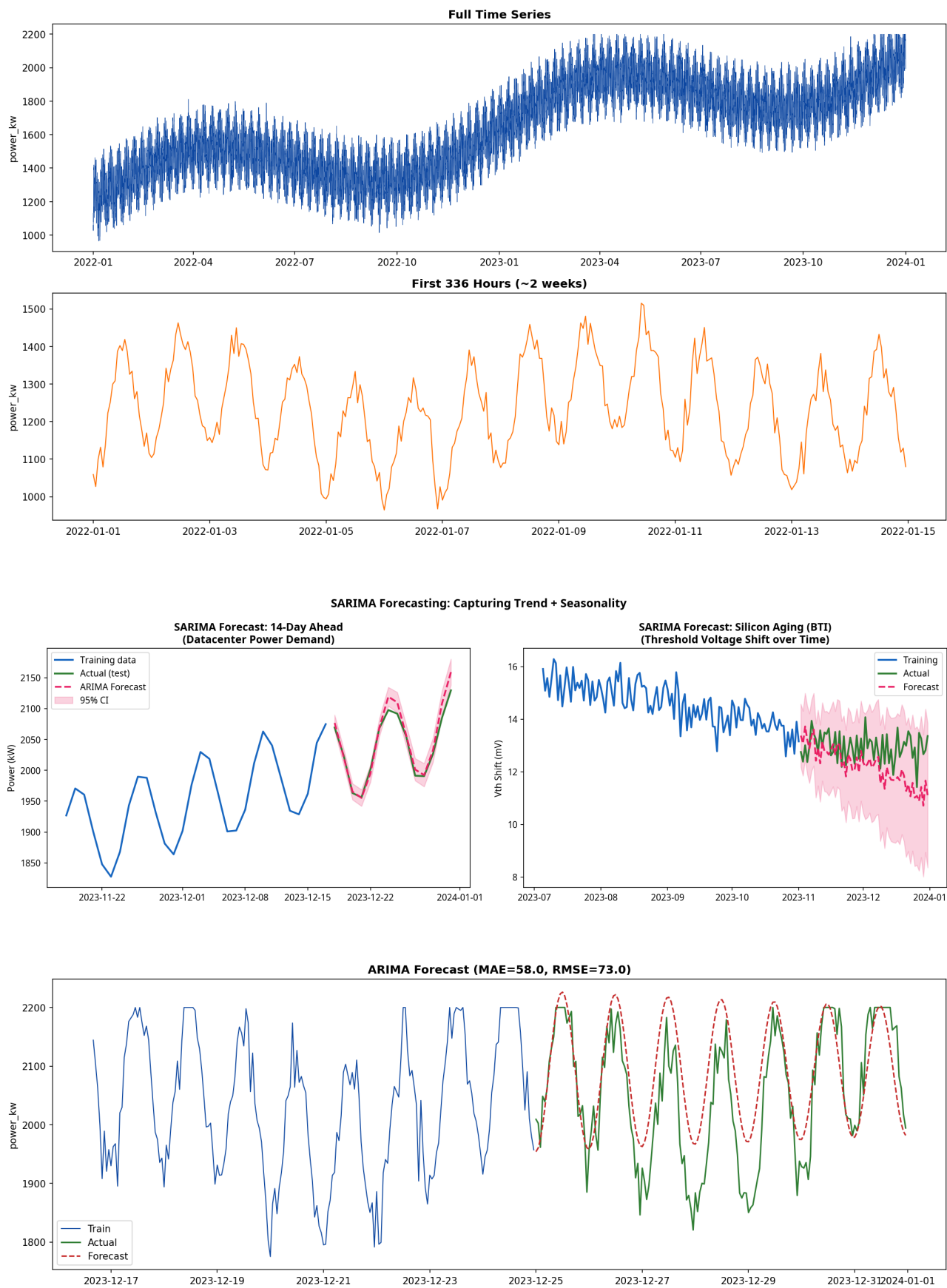
3D Time Series Decomposition Datacenter Power Demand

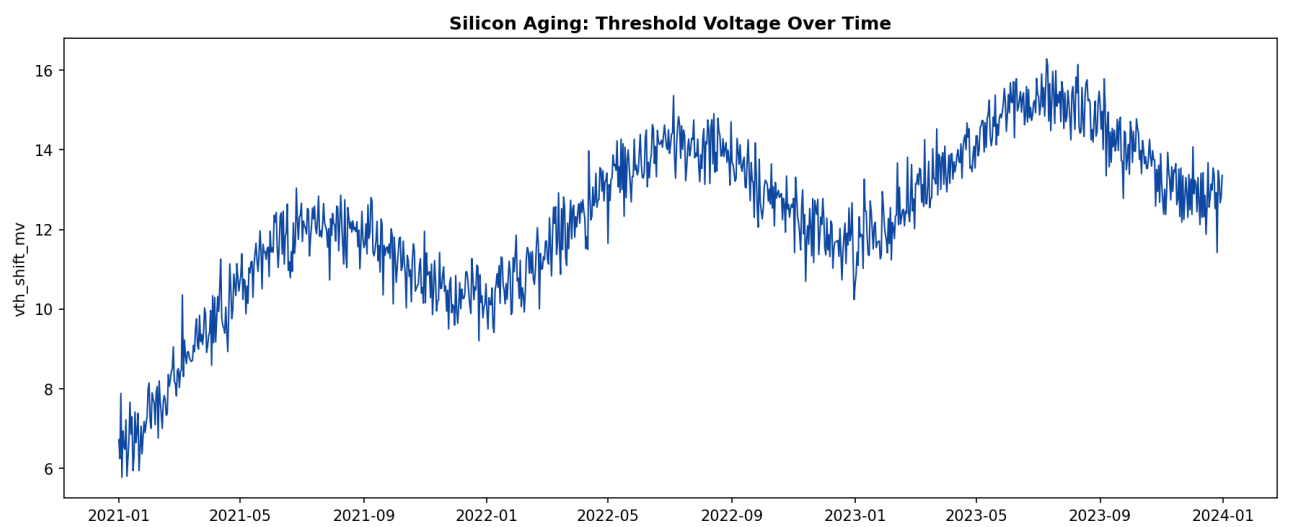
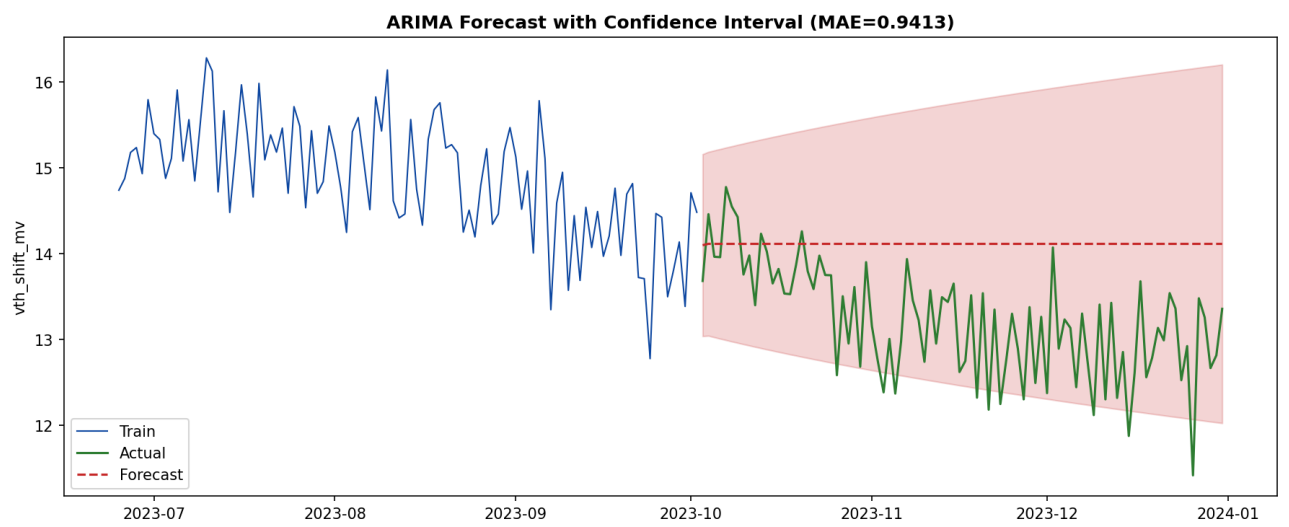
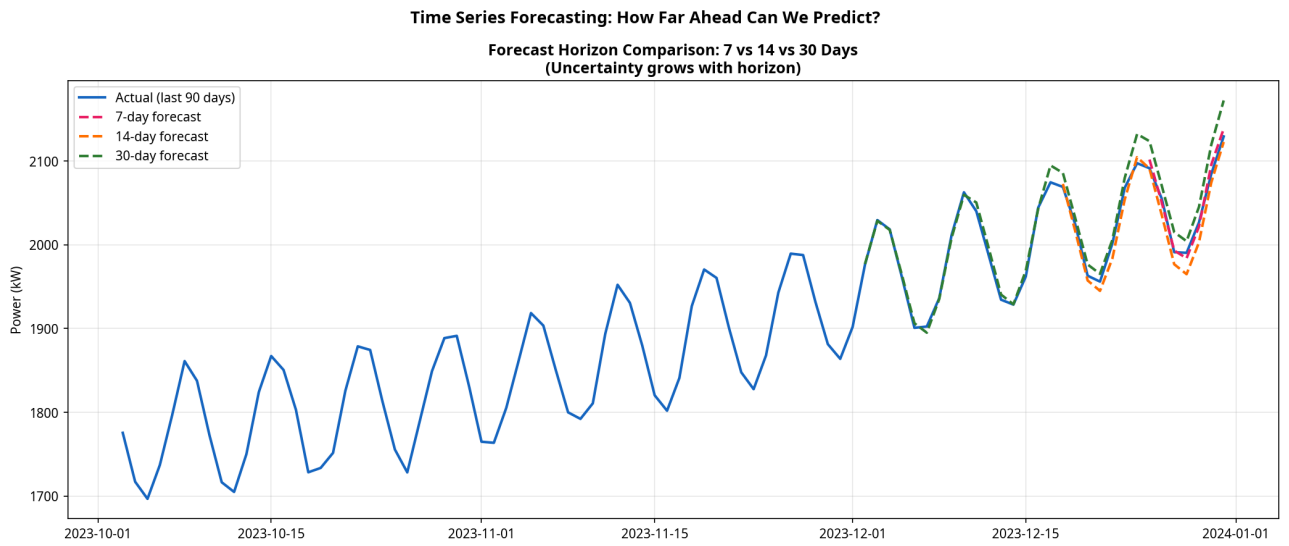
- Trend
- Seasonal
- Residual



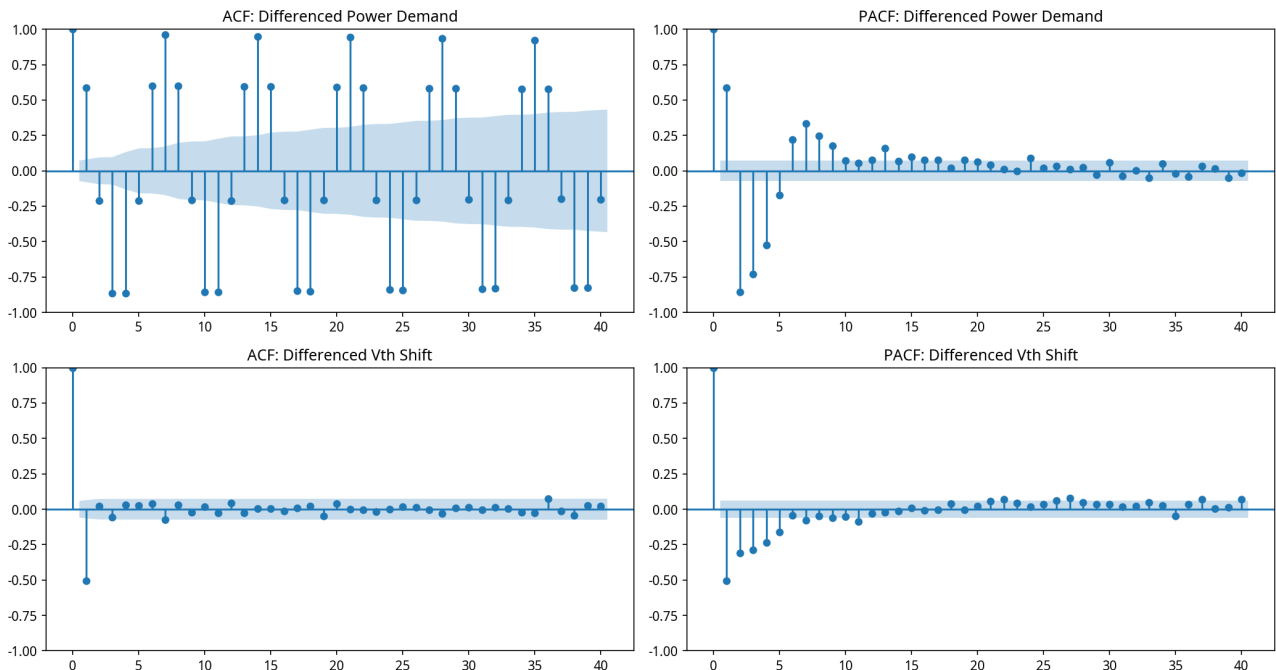




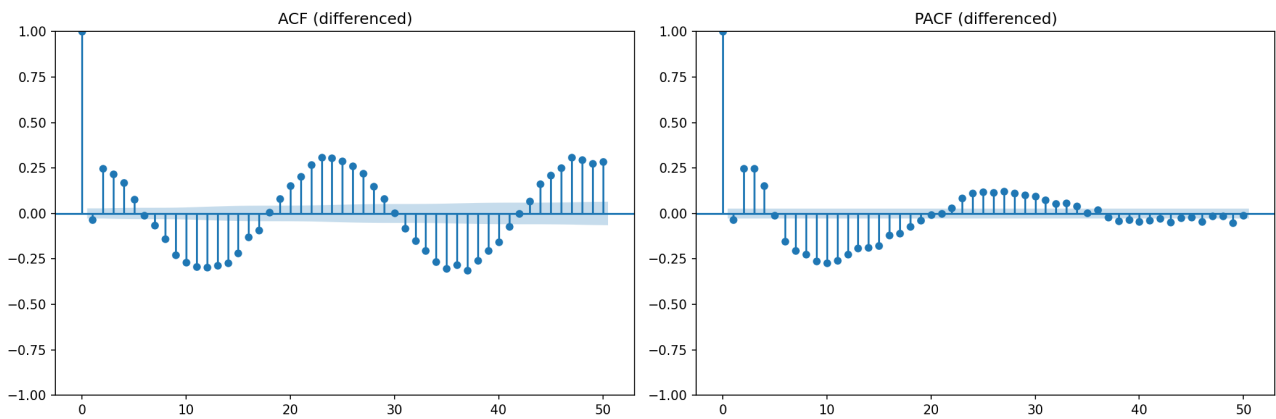




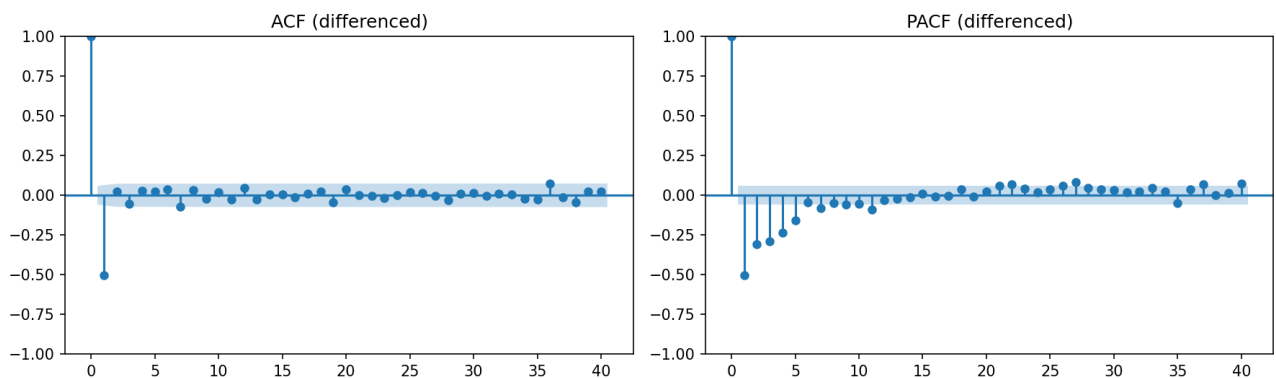
ACF and PACF: Identifying ARIMA Parameters (p, d, q)
(Significant lags tell you the AR and MA order)



ACF/PACF for ARIMA Order Selection



ACF/PACF Analysis



Visualizations showcase trends, seasonality, and the effectiveness of ARIMA/SARIMA forecasts on the datasets.

5 Tech Stack & Key Learnings

Info	Detail
GitHub	https://github.com/AIML-Engineering-Lab/009_time_series_forecasting
Tech Stack	Python 3.11, scikit-learn, pandas, Matplotlib
Licence	MIT

- Differencing is essential for achieving stationarity.
- ACF/PACF plots are crucial for model selection.
- SARIMA effectively manages seasonality in data.